

BUSINESS REQUIREMENTS DOCUMENT (BRD)

P&C Insurance - Certificate of Insurance (COI) Self-Service Generator

Project Information	Details
Project Name	Certificate of Insurance (COI) Self-Service Generator
Business Domain	Property & Casualty (P&C) Insurance
Project Owner	Durja Syamala Pindi
Version	1.0
Target Duration	2 Weeks
Status	Draft - for Review

1. Executive Summary

The Certificate of Insurance (COI) Self-Service Generator is a P&C Insurance solution designed to simplify and digitize the process of obtaining a Certificate of Insurance.

The solution will allow an authorized policyholder's business contact, such as a contractor or business representative, to select an eligible insurance policy, provide the name of the additional-insured party, and generate a Certificate of Insurance as a PDF.

The generated certificate will be produced synchronously and made available to the requester through an on-screen PDF preview and download option. Each successfully generated COI will also be recorded in an audit log to provide traceability.

The Version 1 solution is intentionally focused on a simple self-service experience. It will support one policy per request and will not include email delivery or asynchronous processing.

Expected Business Outcomes

- Reduce manual effort involved in COI requests.
- Reduce dependency on insurance service personnel for routine requests.
- Provide a faster self-service experience to authorized requesters.
- Standardize the COI request and generation process.
- Reduce errors caused by manual data entry.
- Provide an audit trail for generated certificates.
- Improve visibility and traceability of COI generation requests.

2. Business Problem

Obtaining a Certificate of Insurance can involve manual interaction between a policyholder's business contact and the insurance organization. The requester may need to contact an insurance representative, provide policy information and additional-insured details, and wait for the certificate to be prepared and delivered.

For routine COI requests, this creates unnecessary operational effort and introduces dependency on insurance personnel. Manual handling can also result in delays, repeated communication and data-entry errors.

The proposed COI Self-Service Generator addresses this problem by providing an authorized requester with a guided digital process to select a policy, enter the additional-insured party name, generate the certificate and download it immediately.

3. Business Objectives

The primary objectives of the COI Self-Service Generator are:

Enable self-service COI requests for authorized policyholder business contacts.

Reduce manual intervention for routine COI generation.

Provide a simple guided request experience.

Validate required request information before generation.

Generate the COI PDF synchronously.

Provide immediate on-screen PDF preview and download.

Maintain an audit record for every generated COI.

Reduce errors associated with manual data entry.

Standardize the COI generation process.

Provide a foundation that can be extended with additional capabilities in future versions.

4. Scope

4.1 In Scope

- Policy selection.
- Additional-insured party name entry.
- Request validation.
- COI PDF generation.
- PDF preview.
- PDF download.
- Request/audit logging.
- Policy lookup API.
- COI generation API.
- Cloud storage for generated PDFs.
- Guided agent/slot-filling experience.
- Basic validation and error handling.

4.2 Out of Scope

- Email delivery of COI.
- Multiple policies in a single request.
- Asynchronous/background generation.
- Complex policy modification.
- Insurance underwriting decisions.
- Payment processing.
- Manual approval workflow.
- Advanced claims functionality.
- Mobile application.

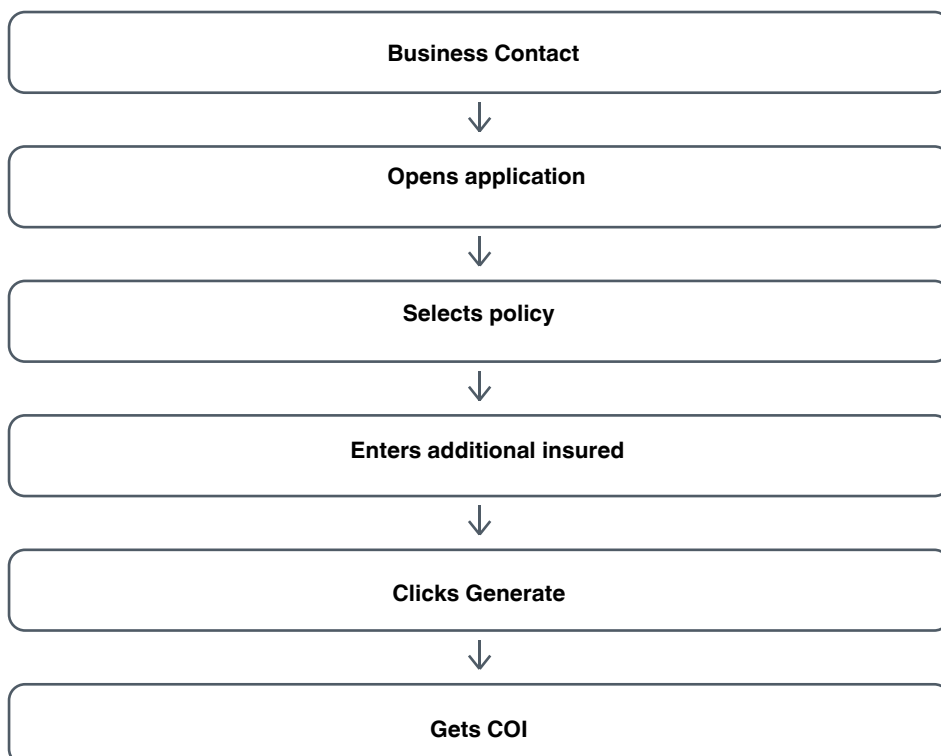
5. Self-Service Concept

Self-service means the authorized requester can complete a routine COI request directly through the application without requiring an insurance employee to perform each step.

Traditional / Manual Process



Self-Service Process



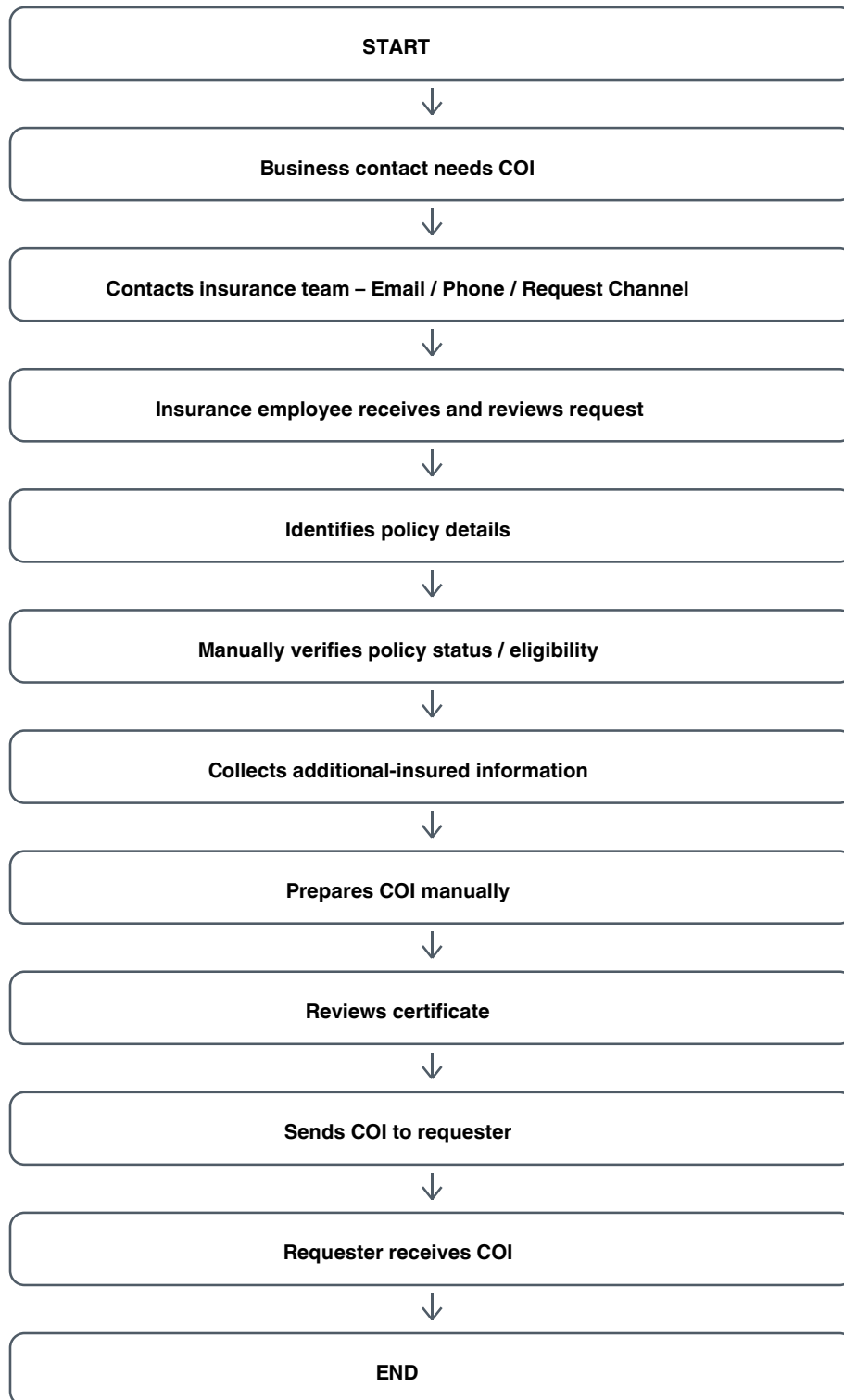
6. Stakeholders / Actors

Stakeholder / Actor	Responsibility
Policyholder's Business Contact / Requester	Selects policy, enters additional-insured name, generates and downloads COI.
P&C; Insurance Business SME	Validates business rules, current-state assumptions, COI content and process.
Business Analyst	Defines requirements, user stories and acceptance criteria.
UX	Designs the short guided form and PDF preview.
Frontend Engineer	Builds React request form and preview/download flow.
API / Backend Engineer	Implements policy lookup, validation and COI generation.
Database	Stores policy/request/audit information.
QA / Test	Validates functional and exception scenarios.
Cloud / DevOps	Supports Cloud Run and Cloud Storage.
Agent	Guides requester through required information.

7. AS-IS Process

Proposed / Assumed AS-IS – To be validated with Business SME. The following process is a preliminary representation derived from the assigned use case and is not a confirmed company process.

AS-IS Process



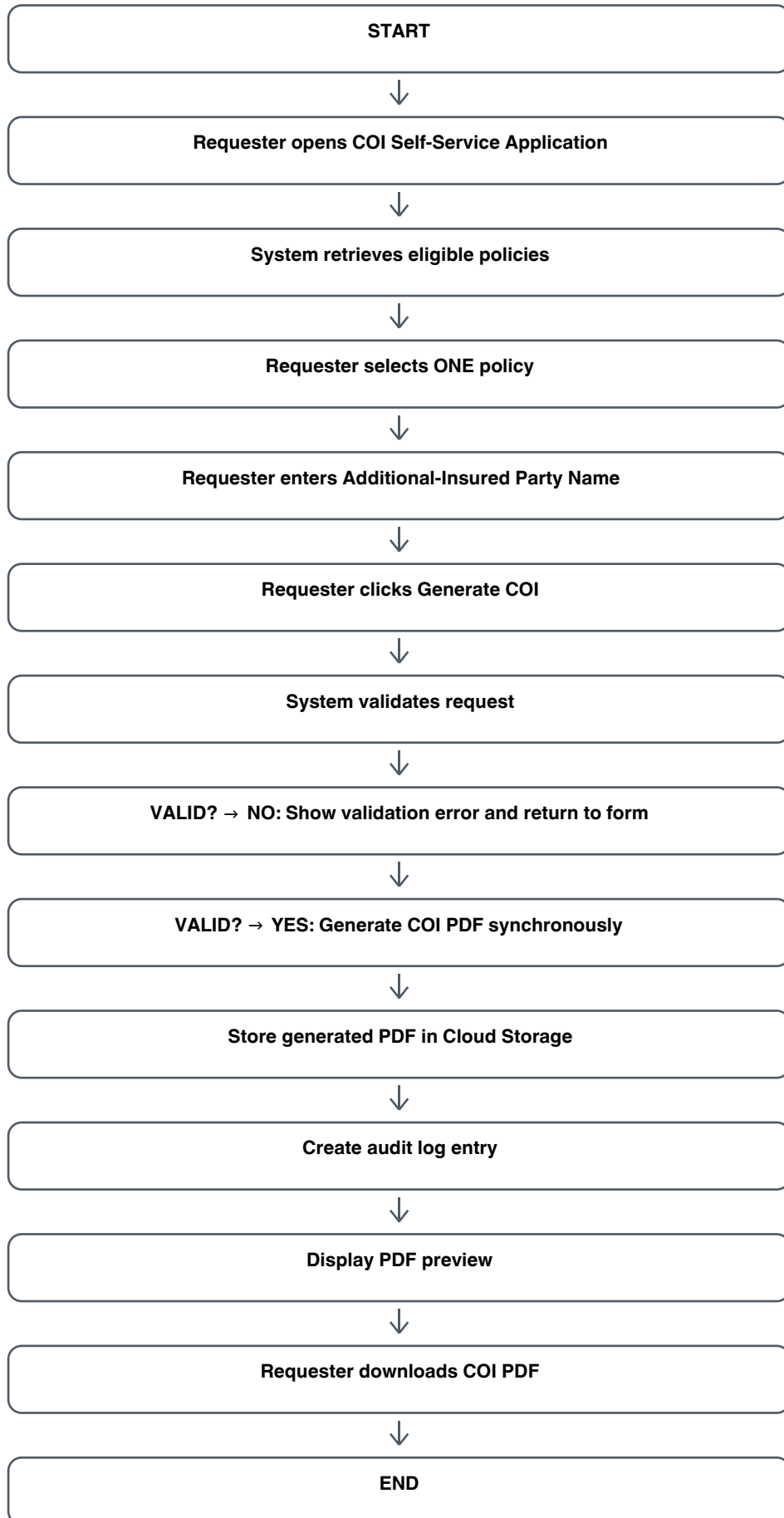
7.2 AS-IS Pain Points

- Employee dependency.
- Manual policy lookup / verification.
- Manual information collection.
- Manual certificate preparation.
- Request turnaround time.
- Repetitive operational effort.
- Potential data-entry errors.
- Manual request tracking.

These are potential pain points and should be validated with the relevant Business SME.

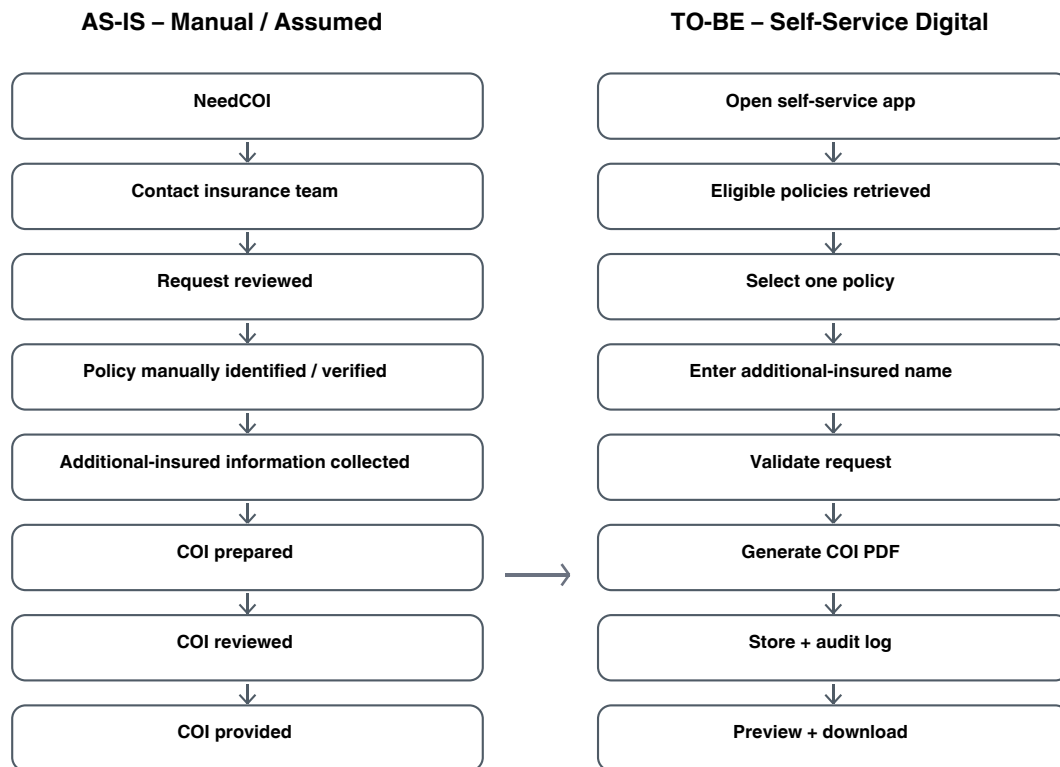
8. Future-State / TO-BE Process

The TO-BE process represents the proposed V1 self-service solution.



9. AS-IS → TO-BE Transformation

The transformation below shows how the proposed solution changes the assumed manual process into a self-service digital process.



Key Improvements

- Manual request → Self-service request
- Manual policy lookup → Automated policy lookup
- Manual data collection → Guided form / Agent
- Manual generation → Automated PDF generation
- Manual tracking → Automated audit logging
- Delayed document receipt → Immediate on-screen availability
- High employee dependency → Reduced employee dependency

10. Business Requirements

	Business Requirement
ID BR-001	The system shall allow an authorized requester to select one eligible policy for a COI request.
BR-002	The system shall allow the requester to provide the name of the additional-insured party.
BR-003	The system shall validate that all mandatory information has been provided before COI generation. The
BR-004	system shall generate a COI PDF for a valid request.
BR-005	The system shall make the generated COI available to the requester for on-screen preview and download.
BR-006	The system shall record every successfully generated COI request for audit purposes.
BR-007	The system shall provide guided assistance to help the requester complete the required information.

11. Functional Requirements

Business requirement: **BR-004: Generate COI**

ID	Functional Requirement	Priority
FR-001	Display eligible policies to requester	Must Have
FR-002	Allow selection of exactly one policy	Must Have
FR-003	Capture additional-insured party name	Must Have
FR-004	Validate mandatory fields	Must Have
FR-005	Validate selected policy	Must Have
FR-006	Generate COI PDF	Must Have
FR-007	Store generated PDF	Must Have
FR-008	Display PDF preview	Must Have
FR-009	Allow PDF download	Must Have
FR-010	Create audit log	Must Have
FR-011	Display meaningful errors	Must Have
FR-012	Support guided agent completion	Should Have

12. Non-Functional Requirements

Area	Requirement
Performance	COI generation should complete within an acceptable user-facing response time for V1.
Availability	Application and required services should be available during supported operating periods.
Security	Only authorized users should access eligible policy information and generated documents.
Auditability	Successful generations must have traceable request records.
Usability	The form should be simple for a non-expert user.
Reliability	Generation/storage failures must not produce a false success.
Scalability	The design should support future increases in COI request volume.

13. Business Rules

BRULE-001 — OneCOI request shall contain exactly one policy.

BRULE-002 — A policy must be eligible/valid before a COI can be generated.

BRULE-003 — Additional-insured party name is mandatory.

BRULE-004 — COI generation shall not occur when mandatory request information is missing.

BRULE-005 — Every successfully generated COI shall have an associated audit record.

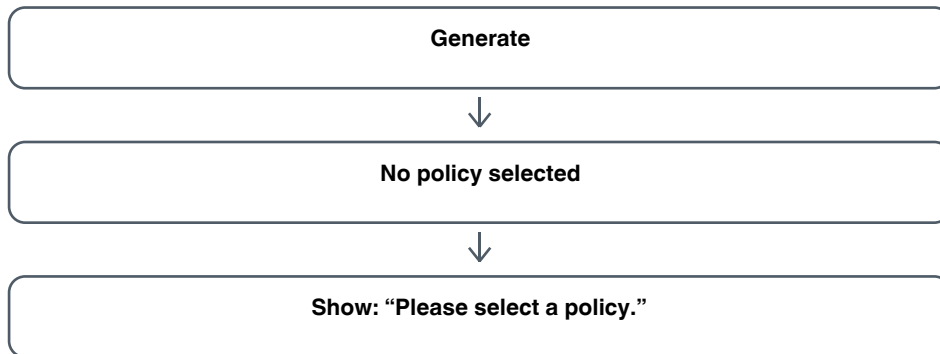
BRULE-006 — Version 1 shall not send the generated COI through email.

BRULE-007 — The generated certificate shall be available through the application for preview/download.

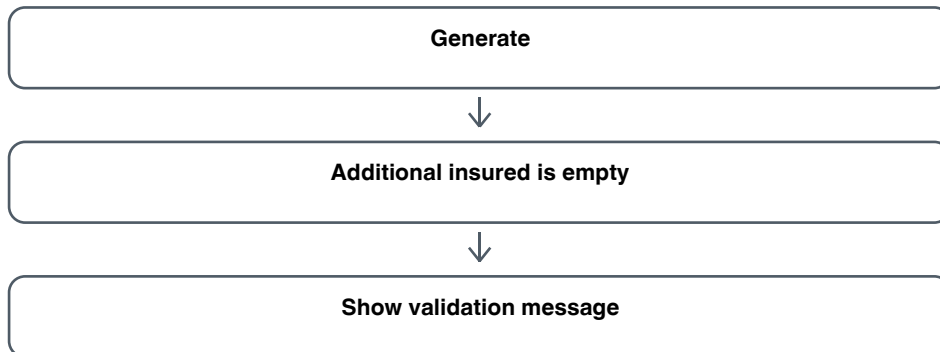
14. Exception Scenarios

A professional BRD should also describe what can go wrong.

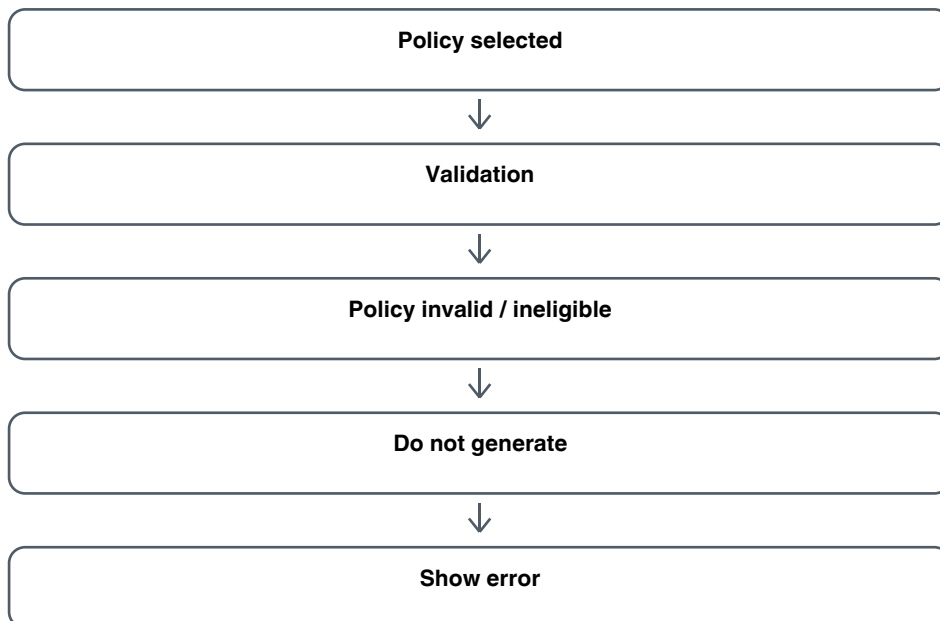
Case 1 — No policy selected



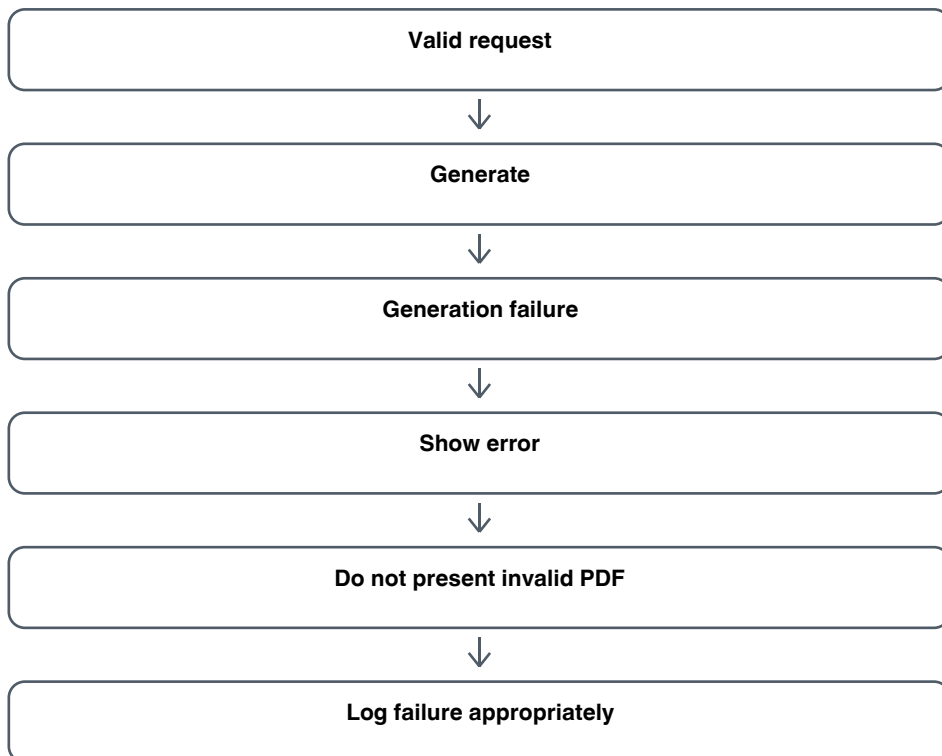
Case 2 — Additional insured missing



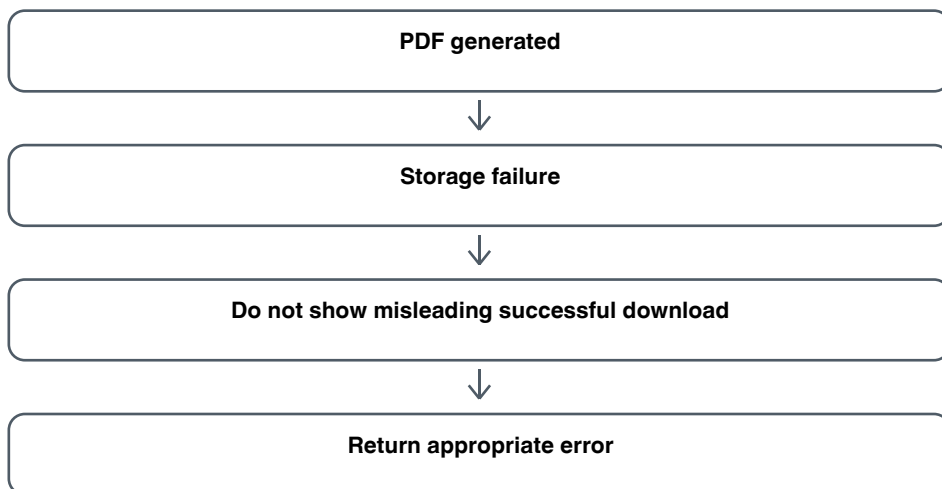
Case 3 — Policy not eligible



Case 4 — PDF generation failure



Case 5 — Storage failure



15. Data Requirements

Data Object	Key Fields
Policy	policyId, policyNumber, policyholder, policyType, effectiveDate, expirationDate, status
COI Request / Audit	requestId, policyId, additionalInsuredName, requestedAt, status, pdfReference

16. Integration Requirements

Component	Purpose
Policy Lookup API / Policy Service	Retrieve policies the requester is allowed to use.
COI Generation API / Service	Generate the COI PDF.
Cloud Run	Run the application/backend services required for V1.
Cloud Storage	Store generated COI PDFs.
Database	Store policy/request/audit information.

17. Roles & Responsibilities

Role	Responsibility
UX	COI request form and PDF preview wireframes.
BA	Requirements, user stories and acceptance criteria.
FE	React request form and preview/download.
API	Policy lookup and COI generation endpoint.
DB	Policies, COI requests and audit data.
Test	Valid request and exception testing.
Cloud	Cloud Run and Cloud Storage.
Agent	Guided slot-filling request.

18. Assumptions

- Therequesteris authorized to access the applicable policy information.
- Policy information is available electronically.
- An approved COI template and required content rules are provided by the business.
- Additional-insured rules are confirmed by the Business SME.
- Cloud Run and Cloud Storage are available for V1.
- V1 supports one policy per request.
- V1 uses synchronous generation.
- V1 does not include email delivery.

19. Dependencies

- Polycysource/service.
- COI PDF generation capability and approved template.
- Cloud Run environment.
- Cloud Storage configuration.
- Database/audit storage.
- Authentication/authorization where required.
- Business SME confirmation of COI rules and current-state process.

20. Risks

Risk	Mitigation / Consideration
Wrong policy selected	Display clear policy information and validate eligibility.
Missing information	Use mandatory validation and Agent guidance.
PDF generation failure	Handle errors and avoid false-success status.
Storage failure	Confirm storage before reporting success.
Incorrect COI template/content	Use SME-approved template and rules.
Unclear AS-IS	Validate assumptions with Business SME.

21. Success Criteria / KPIs

Metric	Success Condition
Valid COI generation	A correct COI PDF is generated and downloadable for a valid request.
Audit logging	Every generated COI is logged for audit.
Agent testing	Agent successfully completes the guided request in 3 test cases.
Invalid request handling	Invalid/incomplete requests are blocked with appropriate feedback.
User outcome	Requester can preview and download a successfully generated COI.

22. Appendix

22.1 V1 Scope Summary

One policy per request | Synchronous generation | No email delivery | On-screen preview/download | Audit logging required.

22.2 Technical Flow Summary

